13 Some Nonlinear Diophantine Equations

An equation with the restriction that only integer (or sometimes rational) solutions are sought is called a *diophantine equation.* We have already studied a simple type of diophantine equation, namely, linear diophantine equations (Section 3.6). We learned how all solutions in integers of a linear diophantine equation can be found. But what about nonlinear diophantine equations?

It is a deep theorem (beyond the scope of this text) that there is no general method for solving all nonlinear diophantine equations. However, many results have been established about particular nonlinear diophantine equations, as well as certain families of nonlinear diophantine equations. This chapter addresses several types of nonlinear diophantine equations. First, we will consider the diophantine equation $x^2 + y^2 = z^2$, satisfied by the lengths of the sides of a right triangle. A triple of integers (x, y, z) that solves this equation is called a Pythagorean triple. After finding an explicit formula for Pythagorean triples, we will show this formula can be found by determining all the points (x, y) on the unit circle with rational coefficients using geometric reasoning.

After studying the diophantine equation $x^2 + y^2 = z^2$, we will consider the famous diophantine equation $x^n + z^n = z^n$, where n is an integer greater than 2. That is, we will be interested in whether the sum of the nth powers of two integers can also be the nth power of an integer, where none of the three integers equals 0. Fermat stated that there are no solutions of this diophantine equation when $n > 2$ (a statement known as Fermat's last theorem), but for more than 350 years no one could find a proof. The first proof of this theorem was discovered by Andrew Wiles in 1995, which ended one of the greatest challenges of mathematics. The proof of Fermat's last theorem is far beyond the scope of this book, but we will be able to provide a proof for the case when $n = 4$.

Next, we will consider the problem of representing integers as the sums of squares. We will determine which integers can be written as the sum of two squares. Furthermore, we will prove that every positive integer is the sum of four squares.

We will also study the diophantine equation $x^2 - dy^2 = 1$, known as Pell's equation. We will show that the solutions of this equation can be found using the simple continued fraction of $\sqrt{d}$, providing another example of the usefulness of continued fractions.

Finally, we will study the famous *congruent number problem,* which asks which integers are the area of a right triangle with sides of integer length. Progress on this ancient problem has been made in recent years through the use of elliptic curves, a type of cubic diophantine equation. We will show how finding rational points on certain elliptic curves can be used to study the congruent number problem.

13.1 Pythagorean Triples

The Pythagorean theorem tells us that the sum of the squares of the lengths of the legs of a right triangle equals the square of the length of the hypotenuse. Conversely, any triangle for which the sum of the squares of the lengths of the two shortest sides equals the square of the third side is a right triangle. Consequently, to find all right triangles with integral side lengths, we need to find all triples of positive integers (x, y, z) satisfying the diophantine equation

$$x^2 + y^2 = z^2. \tag{13.1}$$

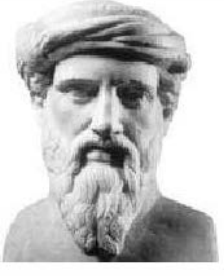
Triples of positive integers satisfying this equation are called *Pythagorean triples* after the ancient Greek mathematician *Pythagoras*. Similarly, we call a right triangle with integer side lengths a *Pythagorean triangle*.

Example 13.1. The triples (3, 4, 5), (6, 8, 10), and (5, 12, 13) are Pythagorean triples because $3^2 + 4^2 = 5^2$, $6^2 + 8^2 = 10^2$, and $5^2 + 12^2 = 13^2$. ◀

Unlike most nonlinear diophantine equations, it is possible to explicitly describe all the integral solutions of (13.1). Before developing the result describing all Pythagorean triples, we need a definition.

Definition. A Pythagorean triple (x, y, z) is called *primitive* if x, y, and z are relatively prime, that is, if $(x, y, z) = 1$. We call a triangle a *primitive right triangle* if its sides have lengths from a primitive Pythagorean triple.

Remark. Unfortunately, the notation (x, y, z) can denote the ordered triple of numbers x, y, and z or the greatest common divisor of x, y, and z. Fortunately, the context in which this notation is used will always make it clear which meaning is intended.



PYTHAGORAS (c. 572–c. 500 B.C.E.) was born on the Greek island of Samos. After extensive travels and studies, Pythagoras founded his famous school at the Greek port of Crotona, in what is now southern Italy. Besides being an academy devoted to the study of mathematics, philosophy, and science, the school was the site of a brotherhood sharing secret rites. The Pythagoreans, as the members of this brotherhood were called, published nothing and ascribed all their discoveries to Pythagoras himself. However, it is believed that Pythagoras himself discovered what is now called the Pythagorean theorem, namely, that $a^2 + b^2 = c^2$, where a, b, and c are the lengths of the two legs and of the hypotenuse of a right triangle, respectively. The Pythagoreans believed that the key to understanding the world lay with natural numbers and form. Their central tenet was "Everything is Number." Because of their fascination with the natural numbers, the Pythagoreans made many discoveries in number theory. In particular, they studied perfect numbers and amicable numbers for the mystical properties they felt these numbers possessed.

Example 13.2. The Pythagorean triples (3, 4, 5) and (5, 12, 13) are primitive, whereas the Pythagorean triple (6, 8, 10) is not. ◀

Let (x, y, z) be a Pythagorean triple with $(x, y, z) = d$. Then there are integers x_1, y_1, z_1 with $x = dx_1$, $y = dy_1$, $z = dz_1$, and $(x_1, y_1, z_1) = 1$. Furthermore, because

$$x^2 + y^2 = z^2,$$

we have

$$(x/d)^2 + (y/d)^2 = (z/d)^2,$$

so that

$$x_1^2 + y_1^2 = z_1^2.$$

Hence, (x_1, y_1, z_1) is a primitive Pythagorean triple, and the original triple (x, y, z) is simply an integral multiple of this primitive Pythagorean triple.

Also note that any integral multiple of a primitive (or for that matter any) Pythagorean triple is again a Pythagorean triple. If (x_1, y_1, z_1) is a primitive Pythagorean triple, then we have

$$x_1^2 + y_1^2 = z_1^2,$$

and hence,

$$(dx_1)^2 + (dy_1)^2 = (dz_1)^2,$$

so that (dx_1, dy_1, dz_1) is a Pythagorean triple.

Consequently, all Pythagorean triples can be found by forming integral multiples of primitive Pythagorean triples. To find all primitive Pythagorean triples, we need some lemmas. The first lemma tells us that any two integers of a primitive Pythagorean triple are relatively prime.

Lemma 13.1. If (x, y, z) is a primitive Pythagorean triple, then $(x, y) = (x, z) = (y, z) = 1$.

Proof. Suppose that (x, y, z) is a primitive Pythagorean triple and $(x, y) > 1$. Then, there is a prime p such that $p \mid (x, y)$, so that $p \mid x$ and $p \mid y$. Because $p \mid x$ and $p \mid y$, we know that $p \mid (x^2 + y^2) = z^2$. Because $p \mid z^2$, we can conclude that $p \mid z$. This is a contradiction, because $(x, y, z) = 1$. Therefore, $(x, y) = 1$. In a similar manner, we can easily show that $(x, z) = (y, z) = 1$. ■

Next, we establish a lemma about the parity of the integers of a primitive Pythagorean triple.

Lemma 13.2. If (x, y, z) is a primitive Pythagorean triple, then x is even and y is odd or x is odd and y is even.

Proof. Let (x, y, z) be a primitive Pythagorean triple. By Lemma 13.1, we know that $(x, y) = 1$, so that x and y cannot both be even. Also, x and y cannot both be odd. If x

and y were both odd, then we would have

$$x^2 \equiv y^2 \equiv 1 \pmod 4,$$

so that

$$z^2 = x^2 + y^2 \equiv 2 \pmod 4.$$

This is impossible. Therefore, x is even and y is odd, or vice versa. ■

The final lemma that we need is a consequence of the fundamental theorem of arithmetic. It tells us that two relatively prime integers that multiply together to give a square must both be squares.

Lemma 13.3. If r, s, and t are positive integers such that $(r, s) = 1$ and $rs = t^2$, then there are integers m and n such that $r = m^2$ and $s = n^2$.

Proof. If $r = 1$ or $s = 1$, then the lemma is obviously true, so we may suppose that $r > 1$ and $s > 1$. Let the prime-power factorizations of r, s, and t be

$$r = p_1^{a_1} p_2^{a_2} \cdots p_u^{a_u},$$
$$s = p_{u+1}^{a_{u+1}} p_{u+2}^{a_{u+2}} \cdots p_v^{a_v},$$

and

$$t = q_1^{b_1} q_2^{b_2} \cdots q_k^{b_k}.$$

Because $(r, s) = 1$, the primes occurring in the factorizations of r and s are distinct. Because $rs = t^2$, we have

$$p_1^{a_1} p_2^{a_2} \cdots p_u^{a_u} p_{u+1}^{a_{u+1}} p_{u+2}^{a_{u+2}} \cdots p_v^{a_v} = q_1^{2b_1} q_2^{2b_2} \cdots q_k^{2b_k}.$$

From the fundamental theorem of arithmetic, the prime powers occurring on the two sides of the above equation are the same. Hence, each p_i must be equal to q_j for some j with matching exponents, so that $a_i = 2b_j$. Consequently, every exponent a_i is even, and therefore $a_i/2$ is an integer. We see that $r = m^2$ and $s = n^2$, where m and n are the integers

$$m = p_1^{a_1/2} p_2^{a_2/2} \cdots p_u^{a_u/2}$$

and

$$n = p_{u+1}^{a_{u+1}/2} p_{u+2}^{a_{u+2}/2} \cdots p_v^{a_v/2}.$$ ■

We can now prove the desired result that describes all primitive Pythagorean triples.

Theorem 13.1. The triple (x, y, z) of positive integers is a primitive Pythagorean triple, with y even, if and only if there are relatively prime positive integers m and n, $m > n$, with m odd and n even or m even and n odd, such that

$$x = m^2 - n^2,$$
$$y = 2mn,$$
$$z = m^2 + n^2.$$

Proof. Let (x, y, z) be a primitive Pythagorean triple. We will show that there are integers m and n as specified in the statement of the theorem. Lemma 13.2 tells us that x is odd and y is even, or vice versa. Because we have assumed that y is even, x and z are both odd. Hence, $z + x$ and $z - x$ are both even, so that there are positive integers r and s with $r = (z + x)/2$ and $s = (z - x)/2$.

Because $x^2 + y^2 = z^2$, we have $y^2 = z^2 - x^2 = (z + x)(z - x)$. Hence,

$$\left(\frac{y}{2}\right)^2 = \left(\frac{z+x}{2}\right)\left(\frac{z-x}{2}\right) = rs.$$

We note that $(r, s) = 1$. To see this, let $(r, s) = d$. Because $d \mid r$ and $d \mid s$, $d \mid (r + s) = z$ and $d \mid (r - s) = x$. This means that $d \mid (x, z) = 1$, so that $d = 1$.

Using Lemma 13.3, we see that there are positive integers m and n such that $r = m^2$ and $s = n^2$. Writing x, y, and z in terms of m and n, we have

$$\begin{aligned} x &= r - s = m^2 - n^2, \\ y &= \sqrt{4rs} = \sqrt{4m^2n^2} = 2mn, \\ z &= r + s = m^2 + n^2. \end{aligned}$$

We also see that $(m, n) = 1$, because any common divisor of m and n must also divide $x = m^2 - n^2$, $y = 2mn$, and $z = m^2 + n^2$, and we know that $(x, y, z) = 1$. We also note that m and n cannot both be odd, for if they were, then x, y, and z would all be even, contradicting the condition $(x, y, z) = 1$. Because $(m, n) = 1$ and m and n cannot both be odd, we see that m is even and n is odd, or vice versa. This shows that every primitive Pythagorean triple has the appropriate form.

To complete the proof, we must show that every triple (x, y, z) with

$$\begin{aligned} x &= m^2 - n^2, \\ y &= 2mn, \\ z &= m^2 + n^2, \end{aligned}$$

where m and n are positive integers $m > n$, $(m, n) = 1$, and $m \not\equiv n \pmod 2$, is a primitive Pythagorean triple. First, note that $m^2 - n^2$, $2mn$, $m^2 + n^2$ forms a Pythagorean triple because

$$\begin{aligned} x^2 + y^2 &= (m^2 - n^2)^2 + (2mn)^2 \\ &= (m^4 - 2m^2n^2 + n^4) + 4m^2n^2 \\ &= m^4 + 2m^2n^2 + n^4 \\ &= (m^2 + n^2)^2 \\ &= z^2. \end{aligned}$$

To see that this triple forms a primitive Pythagorean triple, we must show that these values of x, y, and z are mutually relatively prime. Assume for the sake of contradiction that $(x, y, z) = d > 1$. Then there is a prime $p \mid (x, y, z)$. We note that $p \neq 2$, because x is odd (because $x = m^2 - n^2$, where m^2 and n^2 have opposite parity). Also, note that

because $p \mid x$ and $p \mid z$, $p \mid (z + x) = 2m^2$ and $p \mid (z - x) = 2n^2$. Hence, $p \mid m$ and $p \mid n$, contradicting the fact that $(m, n) = 1$. Therefore, $(x, y, z) = 1$, and (x, y, z) is a primitive Pythagorean triple, concluding the proof. ■

The following example illustrates the use of Theorem 13.1 to produce a Pythagorean triple.

Example 13.3. Let $m = 5$ and $n = 2$, so that $(m, n) = 1$, $m \not\equiv n \pmod 2$, and $m > n$. Hence, Theorem 13.1 tells us that (x, y, z) with

$$\begin{aligned} x &= m^2 - n^2 = 5^2 - 2^2 = 21, \\ y &= 2mn = 2 \cdot 5 \cdot 2 = 20, \\ z &= m^2 + n^2 = 5^2 + 2^2 = 29 \end{aligned}$$

is a primitive Pythagorean triple. ◀

We list the primitive Pythagorean triple generated using Theorem 13.1 with $m \leq 6$ in Table 13.1.

Rational Points on the Unit Circle

We now turn our attention to a problem in *diophantine geometry,* the subject of finding points on algebraic curves whose coordinates are all integers or are all rational numbers. Points with rational coefficients on a curve are called *rational points* on this curve. We will find all rational points on the unit circle $x^2 + y^2 = 1$ using geometric reasoning.

An immediate benefit of finding all rational points on the unit circle is that we can find all Pythagorean triples from these rational points. To see the relationship between Pythagorean triples and rational points on the unit circle, first suppose that a, b, and c are integers with $c \neq 0$ and $a^2 + b^2 = c^2$ (so that (a, b, c) is a Pythagorean triple when these integers are positive). Dividing both sides of this equation by c^2, we obtain

$$(a/c)^2 + (b/c)^2 = 1.$$

m	n	$x = m^2 - n^2$	$y = 2mn$	$z = m^2 + n^2$
2	1	3	4	5
3	2	5	12	13
4	1	15	8	17
4	3	7	24	25
5	2	21	20	29
5	4	9	40	41
6	1	35	12	37
6	5	11	60	61

Table 13.1 *Some primitive Pythagorean triples.*

Hence, the point $(a/c, b/c)$ is a rational point on the unit circle $x^2 + y^2 = 1$, so that every Pythagorean triple has an associated rational point on the unit circle.

Conversely, suppose that the point (x, y) is a rational point on the unit circle, so that $x^2 + y^2 = 1$ where x and y are rational numbers. Because both x and y are rational numbers, we can express each as a ratio of two integers where the denominator is not zero. By choosing the least common denominator for these rational numbers, we can write $x = a/c$ and $y = b/c$ where a, b, and c are integers with $c \neq 0$ and

$$(a/c)^2 + (b/c)^2 = 1.$$

Multiplying both sides by c^2 tells us that $a^2 + b^2 = c^2$. So, if a and b are both positive, then (a, b, c) is a Pythagorean triple.

We now use some simple ideas from geometry to find the rational points on the unit circle. First, note that the points $(0, 1)$, $(0, -1)$, $(1, 0)$, and $(-1, 0)$ are rational points on this circle. Of these four points, we choose the point $(-1, 0)$ to begin our work. Next, observe that if (x, y) is a point with rational coefficients in the plane, then the slope of the line between (x, y) and $(-1, 0)$ is $t = y/(x + 1)$, which is also rational. Now suppose that t is rational number and consider the line $y = t(x + 1)$ that goes through $(-1, 0)$. We will show that this line intersects the unit circle in a second rational point (see Figure 13.1). This will allow us to parameterize all rational points of the unit circle other than $(-1, 0)$ in terms of the rational number t. (In general, the *parameterization* of a curve is the specification of the points on this curve in terms of one or more variables.)

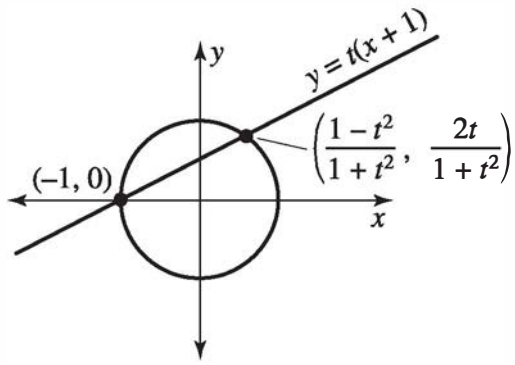


Figure 13.1 *Parameterizing rational points on the unit circle.*

To find the intersection of the line $y = t(x + 1)$ with the unit circle $x^2 + y^2 = 1$, we substitute $t(x + 1)$ for y in the equation for this circle and solve for x. We find that

$$x^2 + t^2(x + 1)^2 = 1.$$

We next subtract 1 from both sides and factor $x^2 - 1$ to obtain

$$(x^2 - 1) + t^2(x + 1)^2 = (x + 1)(x - 1) + t^2(x + 1)^2 = 0.$$

Factoring out the common factor $x + 1$ tells us that

$$(x + 1)[(x - 1) + t^2(x + 1)] = 0.$$

We note that $x = -1$ is a solution; this is no surprise because $(-1, 0)$ is on the line. The other solution is found by solving

$$(x - 1) + t^2(x + 1) = 0$$

for x. This gives $x = (1 - t^2)/(1 + t^2)$. We find the corresponding value for y using the equation of the line $y = t(x + 1)$. This tells us that

$$y = t(x + 1) = t\left(\frac{1 - t^2}{1 + t^2} + 1\right) = t\left(\frac{1 - t^2}{1 + t^2} + \frac{1 + t^2}{1 + t^2}\right) = \frac{2t}{1 + t^2}.$$

We conclude that the second point of intersection of the line $y = t(x + 1)$ with the unit circle is the point $(\frac{1-t^2}{1+t^2}, \frac{2t}{1+t^2})$. This is a rational point when t is rational, because both of its coordinates are rational functions of t (and rational functions of a rational number t are rational because they are the quotient of two polynomials in t, and products, sums, and quotients of rational numbers are rational).

We have found all the rational points on the unit circle, namely, $(-1, 0)$ and all points of the form $(\frac{1-t^2}{1+t^2}, \frac{2t}{1+t^2})$ where t is rational.

When we take $t = m/n$, where m and n are positive integers, in the parameterization we have found for the rational points on the unit circle, we obtain a formula for all Pythagorean triples. That is, given positive integers m and n, we obtain the rational point $(\frac{m^2-n^2}{m^2+n^2}, \frac{2mn}{m^2+n^2})$ on the unit circle. From our earlier comments, we see that $(m^2 - n^2, 2mn, m^2 + n^2)$ is a Pythagorean triple.

Note that when we found the rational points on the unit circle, we found the rational points on an algebraic curve of the form $f(x, y) = 0$ where $f(x, y)$ is a polynomial with integer coefficients. This is an important type of diophantine problem. By expressing the rational points in terms of the rational number t, we gave a rational parameterization of this curve. See Exercises 21–24 for additional examples of rational parameterizations of algebraic curves.

13.1 Exercises

1. a) Find all primitive Pythagorean triples (x, y, z) with $z \leq 40$.
   b) Find all Pythagorean triples (x, y, z) with $z \leq 40$.
2. Show that if (x, y, z) is a primitive Pythagorean triple, then either x or y is divisible by 3.
3. Show that if (x, y, z) is a primitive Pythagorean triple, then exactly one of x, y, and z is divisible by 5.
4. Show that if (x, y, z) is a primitive Pythagorean triple, then at least one of x, y, and z is divisible by 4.
5. Show that every positive integer greater than 2 is part of at least one Pythagorean triple.
6. Let $x_1 = 3$, $y_1 = 4$, $z_1 = 5$, and let x_n, y_n, z_n, for $n = 2, 3, 4, \ldots$, be defined recursively by

$$x_{n+1} = 3x_n + 2z_n + 1,$$
$$y_{n+1} = 3x_n + 2z_n + 2,$$
$$z_{n+1} = 4x_n + 3z_n + 2.$$

Show that (x_n, y_n, z_n) is a Pythagorean triple.

7. Show that if (x, y, z) is a Pythagorean triple with $y = x + 1$, then (x, y, z) is one of the Pythagorean triples given in Exercise 6.

8. Find all solutions in positive integers of the diophantine equation $x^2 + 2y^2 = z^2$.

9. Find all solutions in positive integers of the diophantine equation $x^2 + 3y^2 = z^2$.

* 10. Find all solutions in positive integers of the diophantine equation $w^2 + x^2 + y^2 = z^2$.

11. Find all Pythagorean triples containing the integer 12.

12. Find formulas for the integers of all Pythagorean triples (x, y, z) with $z = y + 1$.

13. Find formulas for the integers of all Pythagorean triples (x, y, z) with $z = y + 2$.

* 14. Show that the number of Pythagorean triples (x, y, z) (with $x^2 + y^2 = z^2$) with a fixed integer x is $(\tau(x^2) - 1)/2$ if x is odd, and $(\tau(x^2/4) - 1)/2$ if x is even.

* 15. Find all solutions in positive integers of the diophantine equation $x^2 + py^2 = z^2$, where p is a prime.

16. Find all solutions in positive integers of the diophantine equation $1/x^2 + 1/y^2 = 1/z^2$.

17. Show that $(f_n f_{n+3}, 2f_{n+1}f_{n+2}, f_{n+1}^2 + f_{n+2}^2)$ is a Pythagorean triple, where f_k denotes the kth Fibonacci number.

18. Find the length of the sides of all right triangles, where the sides have integer lengths and the area equals the perimeter.

19. Find all rational points on the unit circle $x^2 + y^2 = 1$ by determining the intersection of a line with rational slope t that goes through the point $(1, 0)$ with the unit circle.

20. Find all rational points on the unit circle $x^2 + y^2 = 1$ by determining the intersection of a line with rational slope t that goes through $(0, 1)$ with the unit circle.

21. Find all rational points on the circle $x^2 + y^2 = 2$ by determining the intersection of a line with rational slope t that goes through $(1, 1)$ with this circle.

22. Find all rational points on the ellipse $x^2 + 3y^2 = 4$ by determining the intersection of a line with rational slope t that goes through $(1, 1)$ with this ellipse.

23. Find all rational points on the ellipse $x^2 + xy + y^2 = 1$ by determining the intersection of a line with rational slope t that goes through the point $(-1, 0)$ with this ellipse.

24. Suppose that d is a positive integer. Find all rational points on the hyperbola $x^2 - dy^2 = 1$ by determining the intersection of a line with rational slope t that goes through the point $(-1, 0)$ on the hyperbola.

25. Show that there are no rational points on the circle $x^2 + y^2 = 3$.

26. Show that there are no rational points on the circle $x^2 + y^2 = 15$.

* 27. Find all rational points on the unit sphere $x^2 + y^2 + z^2 = 1$. (*Hint:* Use the stereographic projection of the unit sphere to the plane $z = 0$. This projection maps the point (x, y, z) on the sphere to the a point $(u, v, 0)$ that is the intersection of the line through this point and $(0, 0, 1)$, the north pole of the sphere, and the plane $z = 0$. Parameterize the rational

points on the unit sphere using two rational parameters u and v corresponding to this point of intersection.)

Computations and Explorations

1. Find as many Pythagorean triples (x, y, z) as you can, where each of x, y, and z is 1 less than the square of an integer. Do you think that there are infinitely many such triples?
2. Let $\Delta(n)$ denote the number of primitive Pythaogrean triples with hypotenuse less than n. Find $\Delta(10^i)$ for $1 \leq i \leq 6$. By examining $\Delta(10^i)/10^i$ for these values of i, formulate a conjecture for the value approached by $\Delta(n)/n$ as n grows without bound.

Programming Projects

1. Given a positive integer n, find all Pythagorean triples containing n.
2. Given a positive integer n, find all Pythagorean triples with hypotenuse $< n$.
3. Given a positive integer n, find the number of primitive Pythagorean triples with hypotenuse $< n$.

13.2 Fermat's Last Theorem

In the previous section, we showed that the diophantine equation $x^2 + y^2 = z^2$ has infinitely many solutions in nonzero integers x, y, z. What happens when we replace the exponent 2 in this equation with an integer greater than 2? Next to the discussion of the equation $x^2 + y^2 = z^2$ in his copy of the works of Diophantus, Fermat wrote in the margin:

> However, it is impossible to write a cube as the sum of two cubes, a fourth power as the sum of two fourth powers and in general any power as the sum of two similar powers. For this I have discovered a truly wonderful proof, but the margin is too small to contain it.

Fermat did have a proof of this theorem for the special case of $n = 4$. We will present a proof for this case, using his basic methods, later in this section. Although we will never know for certain whether Fermat had a proof of this result for all integers $n > 2$, mathematicians believe it is extremely unlikely that he did. By 1800, all other statements that he made in the margins of his copy of the works of Diophantus were resolved; some were proved and some were shown to be false. Nevertheless, the following theorem is called *Fermat's last theorem.*

Theorem 13.2. ***Fermat's Last Theorem.*** The diophantine equation

$$x^n + y^n = z^n$$

has no solutions in nonzero integers x, y, and z when n is an integer with $n \geq 3$.